



HCMUTE

TRƯỜNG ĐẠI HỌC
SƯ PHẠM KỸ THUẬT TP. HỒ CHÍ MINH
HCMC University of Technology and Education

DRIVER DROWSINESS DETECTION BY USING WEBCAM

Special Project (EL)

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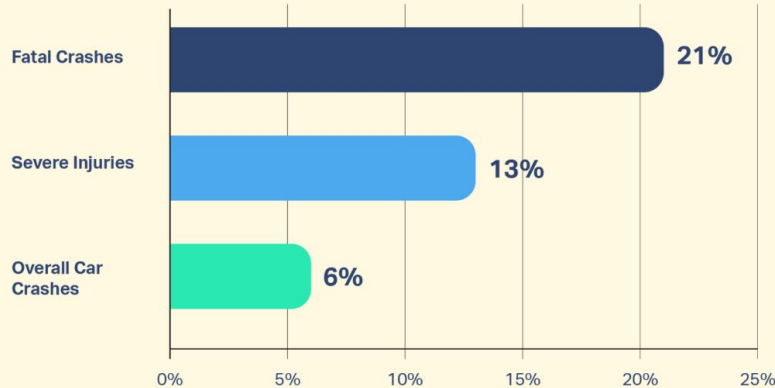
01

REASONS OF CHOOSING THE TOPIC

1. Reason for choosing the topic

1.1. The rising car accident rates due to driver drowsiness

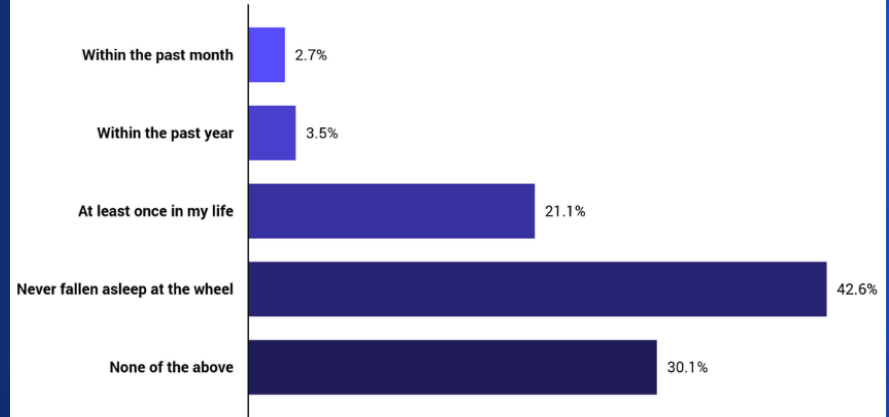
Percentage of U.S. vehicle crashes involving a drowsy driver



Source: AAA Foundation for Traffic Safety

Have you ever fallen asleep, or "nodded off" at the wheel?

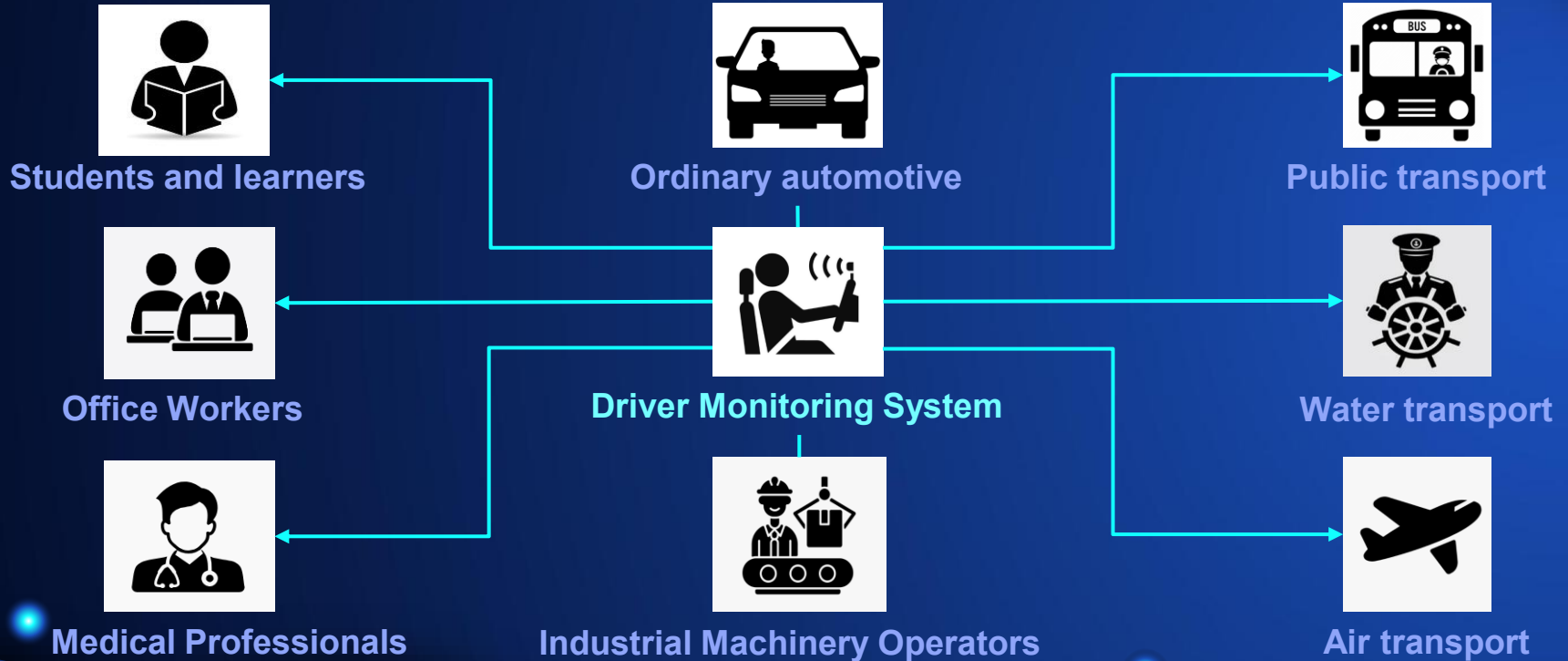
According to a 2019 survey of 2000 Americans



Sources: Annie Atherton, *Drowsy Driving Is a Factor in 21% of Fatal Crashes*. November 6, 2023. The Zebra, *Drowsy Driving Statistics*, January 31, 2023.

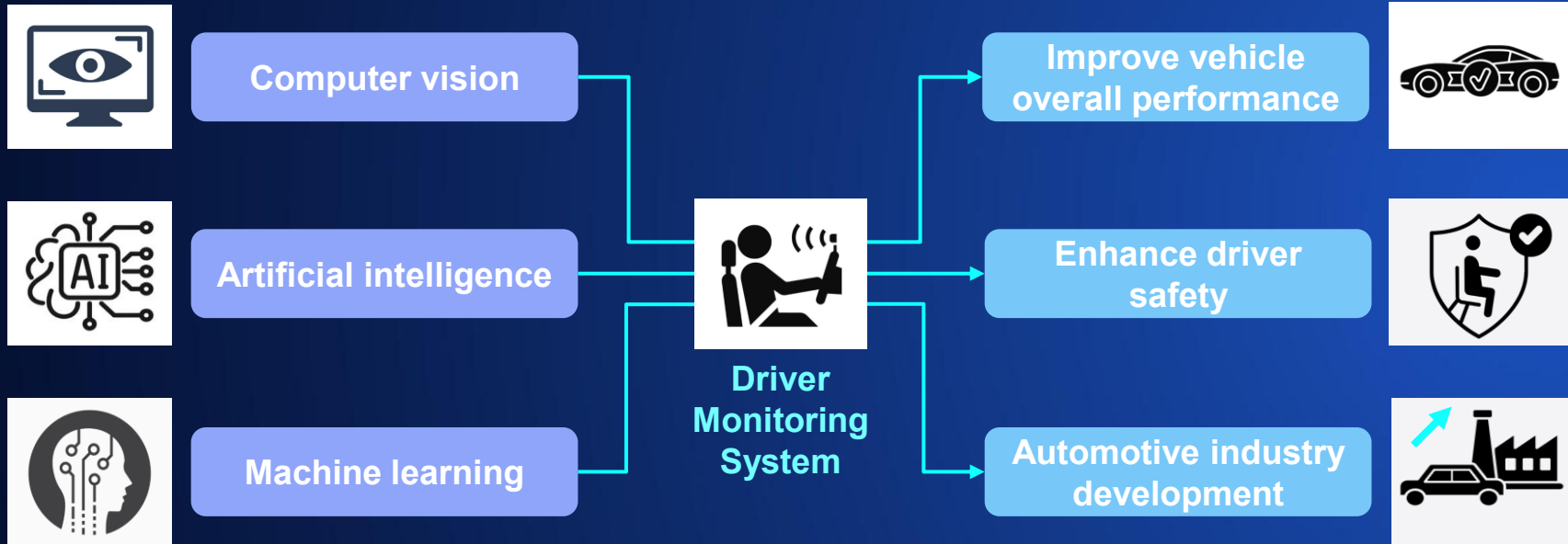
1. Reason for choosing the topic

1.2. Technology applicable across various fields



1. Reason for choosing the topic

1.3. The advancement in automotive technology





02

MAIN COMPONENTS



2. MAIN COMPONENTS



Raspberry Pi 4B

1.500.000 VND



Logitech C270
HD Webcam

200.000 VND

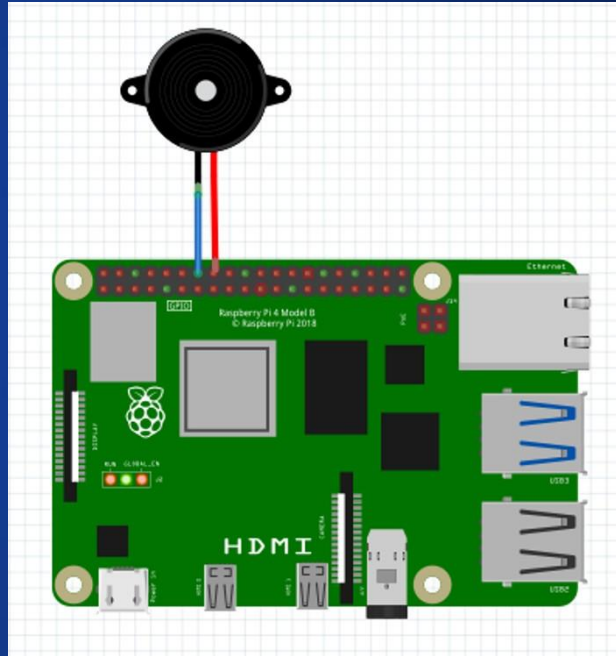


PowerBank
Anker

1.050.000 VND



2. MAIN COMPONENTS



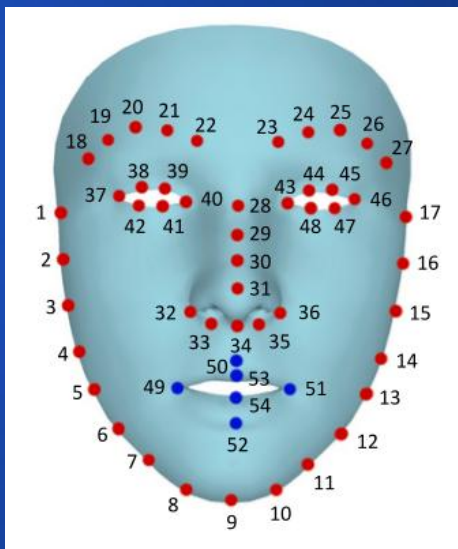


03

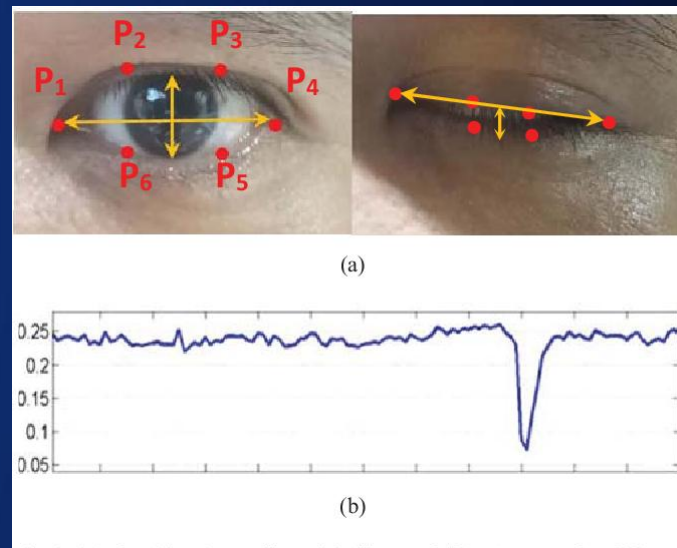
CODING PROCESS

3.1. The theory behind the code

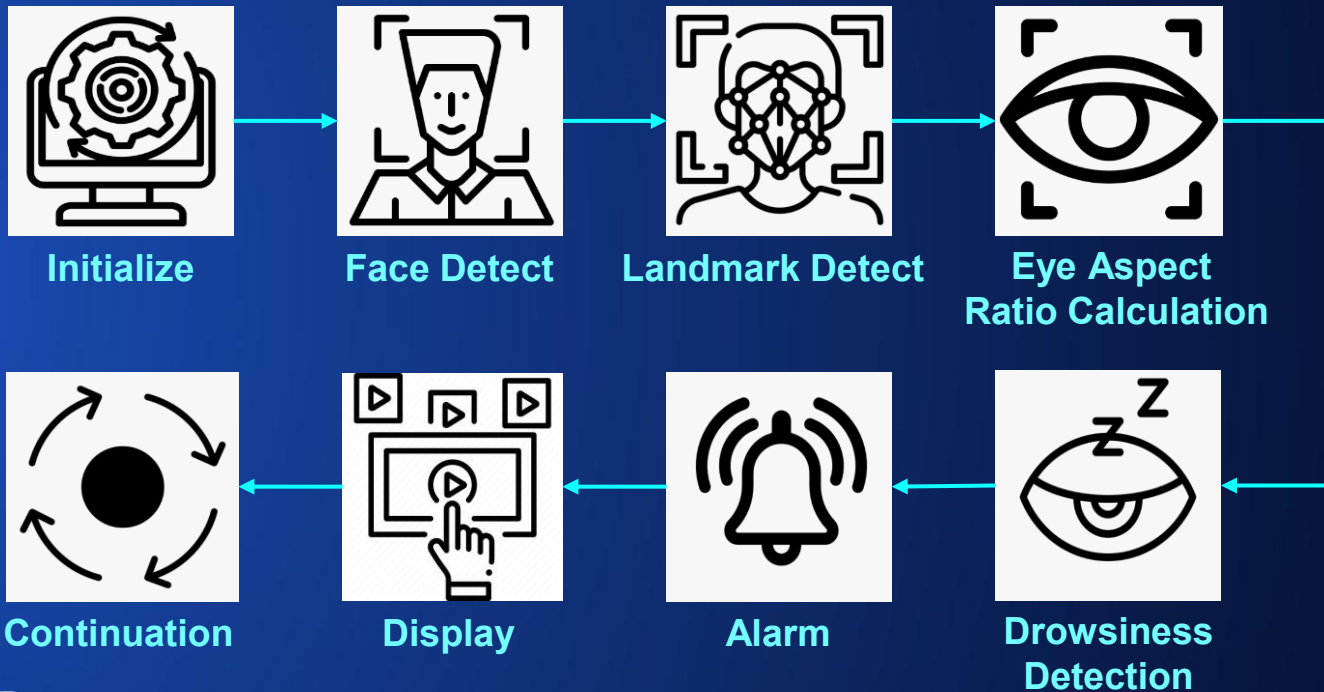
Facial Landmarks



Eye Aspect Ratio (EAR)



3.2. The algorithm behind the code





04

TESTING AND EVALUATIONS

4.1. Testing driver monitoring system

Testing Goals:

Accuracy



Responsiveness



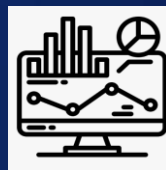
Robustness



Testing Methods:



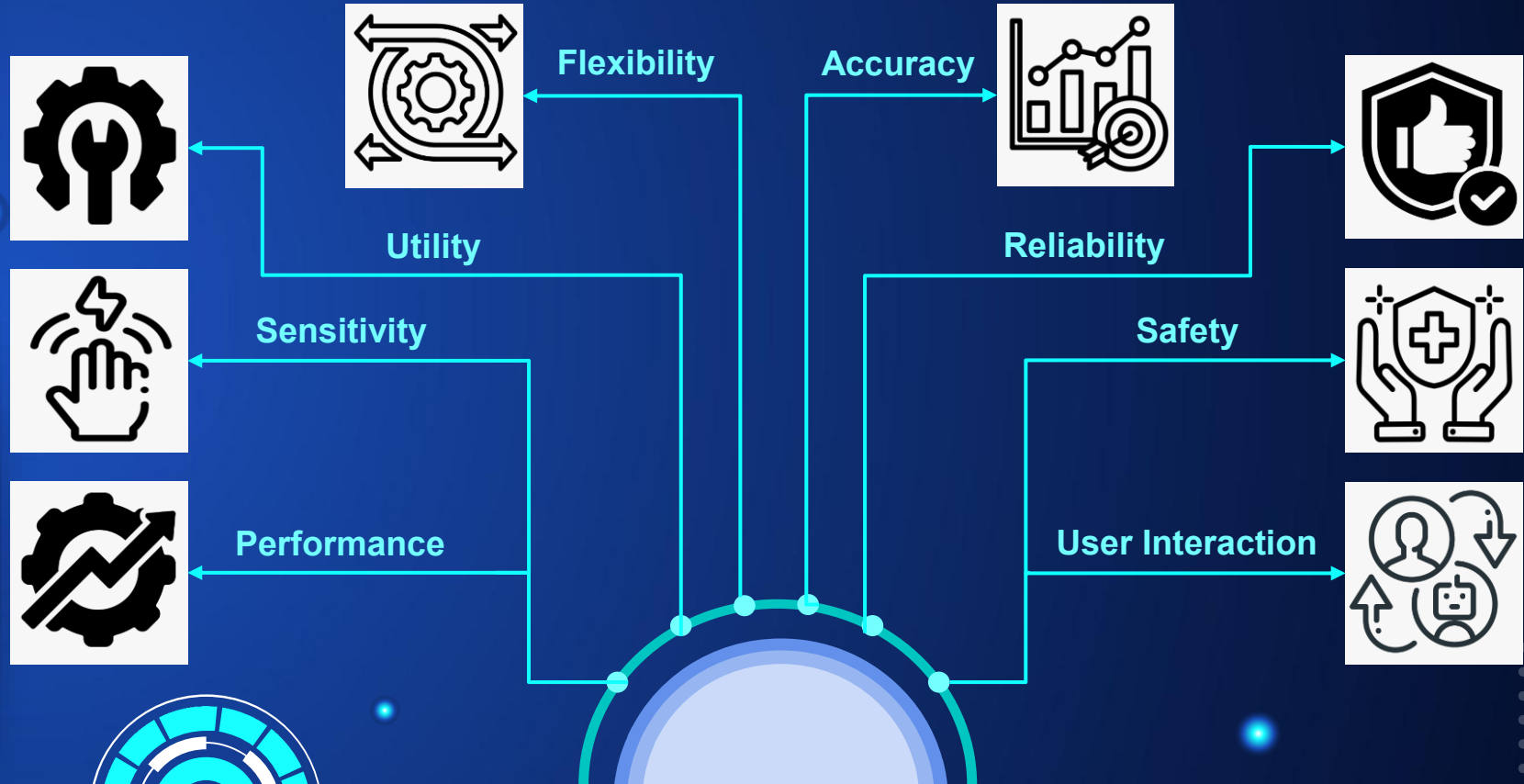
Data Collection



Data Analysis



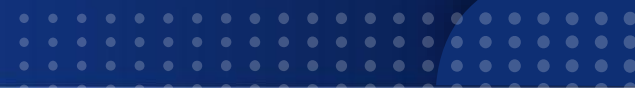
4.2. Driver monitoring system evaluations





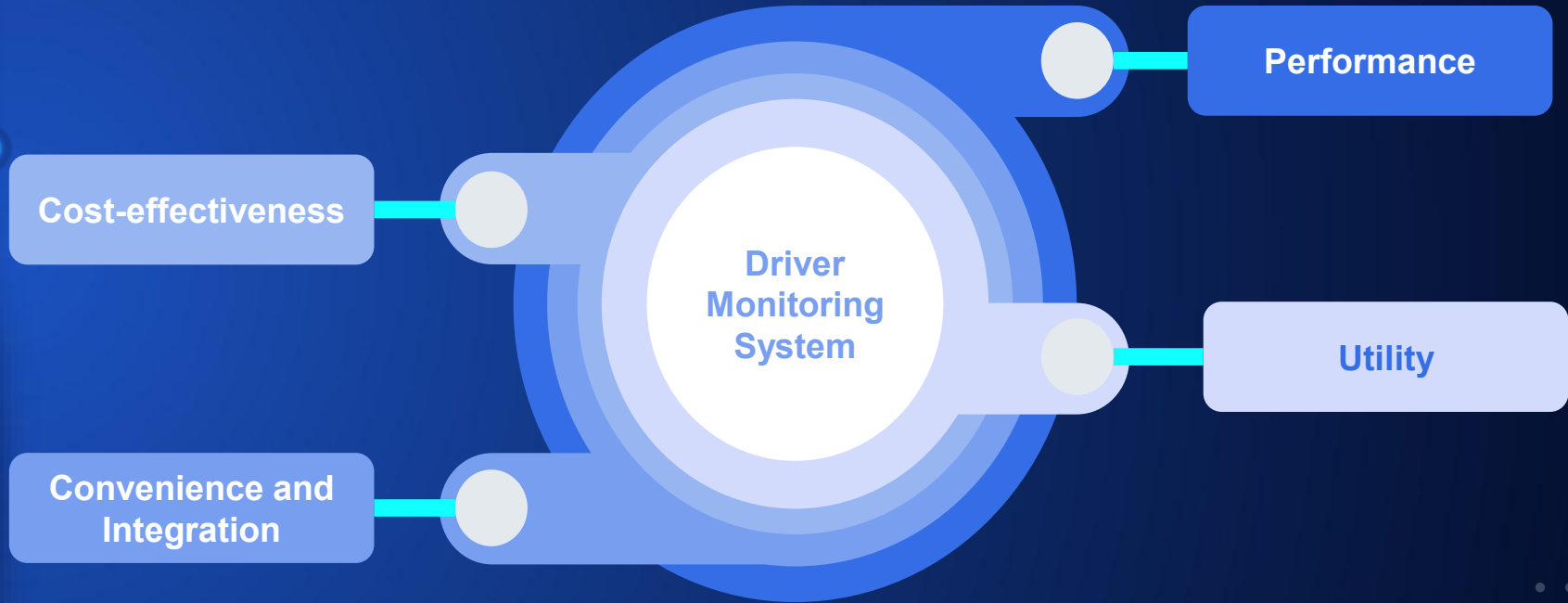
05

CONCLUSION





5. Conclusion





THANKS FOR WATCHING

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